

P2MR Assurance Lab

Independent conformance testing for BIP 360 (Pay-to-Merkle-Root), and the first open measurement of what post-quantum signatures would actually cost inside a Bitcoin transaction — the question BIP 360's own design leaves open.

Program Galaxy Bitcoin Quantum Readiness Initiative

Priority areas #1 implementation/review & #2 post-quantum signature integration **Applicant** Ashwin Goyal · individual · Bhopal, India

License MIT · fully open source

\$55,000

14 weeks, 4 gates, stop rights at each — first decision is \$5,000 for work already public

17–131×

measured on-chain cost of a post-quantum vs. a Schnorr spend — computed, reproducible, on GitHub now

2 real gaps found

independent differential testing already surfaced conformance gaps in the official reference implementation

0 The question nobody has answered — and a first answer, already public

BIP 360 exists so that post-quantum signatures can live in a Bitcoin leaf script “in the future.” Nobody has published what that future actually costs on-chain. This lab computed a first answer — and it is stark.

```
# Reproduce everything below – Python 3.10+, zero dependencies, no network:
git clone https://github.com/let-the-dreamers-rise/p2mr-assurance-lab
cd p2mr-assurance-lab
python verify_vectors.py           # conformance vs official vectors
python differential/run_differential.py  # vs the official reference impl
python pqc/pqc_bench.py           # post-quantum cost measurement

# pqc_bench.py output (exact, from published FIPS 204/205 constants):
a post-quantum P2MR input costs 17x to 131x a Schnorr key-path spend
block input capacity falls from ~17,391 to as few as 133 inputs
9/9 + 7/7 official vectors PASS · 27/27 mutations PASS · 2000/2000 differential trees AGREE
```

That capacity collapse is the crux of Bitcoin's quantum transition, and until now it has been asserted in prose rather than measured. The number above is arithmetic over the standardized FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) signature sizes and Bitcoin's BIP 141 weight model — fully reproducible, with the constants and their sources in `pqc/PARAMS.md`. It is the seed of this proposal's research deliverable, and it is already in the repository.

1 Two things this lab produces that did not exist before

This is not “me too” tooling. It is new knowledge about the leading quantum-resistant Bitcoin proposal — the kind Galaxy's research pillar exists to fund and publish.

NEW CONTRIBUTION	WHY IT IS NEW, AND WHY IT MATTERS TO GALAXY
The first open post-quantum cost model for Bitcoin script	What ML-DSA and SLH-DSA signatures cost as real witness data — vbytes, fee at market rates, and block-capacity impact — has not been published as a rigorous, reproducible Bitcoin-specific analysis. It directly answers whether BIP 360's "post-quantum signatures later" premise is economically practical, which is priority area #2 of Galaxy's own program.
The first independent differential audit of the BIP 360 reference implementation	An independent from-scratch implementation, cross-checked against the official reference on 16 vectors and 2000 random trees, already surfaced two real conformance/robustness gaps (below). No such independent analysis was public before. This is priority area #1, with findings rather than intentions.

The two findings, stated honestly (full grading in FINDINGS.md): the reference implementation applies no depth bound, so it can construct a control block of depth 129 — an output consensus must reject as unspendable; and it validates tree structure with assert, which Python's -O flag strips, turning a malformed tree into a silent wrong result rather than an error. Both are low-severity and neither is a consensus break — they are exactly the class of pre-activation gap independent review exists to catch, reported to maintainers before publication.

2 Why this matters now, and why the framing is honest

Galaxy's announcement names a compressing CRQC timeline, final NIST standards since August 2024, and a 2031 federal deadline, against a Bitcoin upgrade process that "can take years." BIP 360 (P2MR), by Hunter Beast, Ethan Heilman, and Isabel Foxen Duke, is the leading concrete step: it removes Taproot's quantum-vulnerable key path while keeping its script tree, giving long-exposure protection now and a home for post-quantum signature opcodes later. The honest boundary — which every artifact this lab produces states verbatim — is that P2MR does not make Bitcoin quantum-safe today; it protects revealed-key long-exposure risk under fresh-key discipline, and nothing more.

The economic argument is what the cost model sharpens. If a post-quantum spend is 17–131× the size of today's, then fee markets, block capacity, and migration sequencing are the real constraints on Bitcoin's quantum transition — not the cryptography alone. Galaxy's research audience of institutional investors, policymakers, and developers needs that number quantified, sourced, and reproducible. That is the report this program delivers.

3 What USD 55,000 delivers

DELIVERABLE	CONCRETELY
1. Post-quantum cost benchmark — full	The starter model extended to every FIPS 204/205 parameter set at realistic P2MR tree depths, plus measured verification CPU time and malformed-input behavior against a pinned liboqs (or equivalent) build — the one part today's model honestly cannot compute without funded infrastructure. Output: size, fee, block-capacity, and validation-cost tables with full provenance.
2. "What post-quantum Bitcoin costs" — the report	A publication-grade, reproducible Galaxy Research-style paper: methodology, the cost model, migration-capacity implications, and the honest limits. Galaxy Research receives pre-publication review and first-publication option.
3. Conformance corpus — 250+ cases	The 27-case launch corpus grown across every construction and validation surface, each case carrying its expected outcome and specification citation — an importable conformance gate for every P2MR implementer.
4. Differential matrix + disclosure	The independent implementation extended and run against the reference, the bitcoinjs-lib branch, and Core regtest patches, each pinned; findings (including the two already found) carried through coordinated disclosure to maintainers with pinned reproducers.
5. Offline exposure-classifier CLI (v1)	A local-only, zero-telemetry tool classifying output scripts by quantum-exposure class — the seed of the custodian migration tooling of a later phase; refuses all secret material, and the refusals are tested.

4 Why fund this — from your side of the table

A three-week-old program's first grants define its reputation. This one serves two of your four priority areas, already produces findings and a headline research number, and hands your research pillar a first-of-kind paper — at a size you can approve without a committee.

- **You are extending running work, not funding a promise.** The cost model, the differential audit, and the conformance suite are public and reproducible today. M0's \$5,000 is retroactive.
- **It is credible precisely because it does not claim to invent cryptography.** The applicant is not proposing a new post-quantum scheme — an unknown doing that would rightly be dismissed. This is rigorous measurement and independent verification of what already exists, which is exactly what an engineer can do well and what your priority areas ask for.
- **The research output feeds the pillar your council runs.** A reproducible cost paper and a findings log give Barry Sanders, Eran Tromer, and Damien Bérubé concrete artifacts to weigh, and give Galaxy Research something citable to publish.
- **Downside is capped and cheap to administer.** Four gates, each reviewable in under an hour against a one-page acceptance sheet mapping every claim to a command, a file, or a diff. Stop at any gate; maximum exposure to a misjudgment about the applicant is one tranche.
- **Diligence is one email.** Forward this to the BIP 360 authors and ask whether an independent cost model and differential audit are useful right now. This proposal is confident enough in the answer to invite the question.

5 The fourteen weeks

M1 (weeks 1–3) — cost model hardened, corpus to 100+

Extend the cost model across all parameter sets and realistic depths; stand up the pinned PQC build infrastructure for the measured-verification work; grow the conformance corpus past 100 cases under a published taxonomy; ship the threat model, disclosure policy, and a three-platform reproducibility report.

M2 (weeks 4–8) — measured verification cost, 250+ corpus, differential matrix

Measure ML-DSA and SLH-DSA verification cost and malformed-input behavior against the pinned build; complete the 250+ case corpus; run the differential matrix across all pinned implementation targets with minimized reproducers; disclose findings to maintainers first.

M3 (weeks 9–14) — the report, the classifier

Deliver the exposure-classifier v1 with refusal tests, then the report: the full cost model, migration-capacity analysis, differential findings and disclosure outcomes, and honest limits — reproducible end-to-end from public artifacts, Galaxy-reviewed before publication. Six-month maintenance begins.

6 Milestones, payments, stop rights

GATE	WEEK	PAYMENT	CUMULATIVE	ACCEPTED AGAINST
M0	—	\$5,000	\$5,000	The already-public baseline: cost model, differential audit + findings, conformance suite, pinned manifests, scope charter. Retroactive — you pay only for work you have already verified.
M1	3	\$12,000	\$17,000	Full cost model across all parameter sets; PQC build infrastructure pinned; 100+ case corpus; threat model; disclosure policy; three-platform reproducibility report.
M2	8	\$18,000	\$35,000	Measured verification cost and malformed-input behavior; 250+ case corpus; differential matrix across all pinned targets; disclosure log for findings.
M3	14	\$20,000	\$55,000	Exposure-classifier v1 with tested refusals; the report, Galaxy-reviewed, reproducible end-to-end; six-month maintenance commitment begins.

Budget: \$40,000 lead engineering and research (14 weeks, full-time, sole owner) · \$9,000 pinned PQC build + CI + fuzzing + benchmark infrastructure · \$6,000 disclosure, release, and publication operations. Three lines, no vague category. Grant funds never touch trading, tokens, custody, marketing, or undisclosed subcontractors. **Stop rights:** accept, reject, pause, resize, or terminate at any gate; nothing advances automatically; a blocked milestone is reported as blocked. **Automatic freeze** on any unpinned result, any unreproducible claim, any fabricated measurement, or any undisclosed security-relevant finding.

7 Claim discipline — enforced, in the repository today

This banner ships verbatim on every artifact this grant funds:
“BIP 360 (P2MR) is a Draft and is not activated on Bitcoin mainnet. P2MR can mitigate long-exposure (revealed-key) risk only under fresh-key discipline; it does not by itself defend the reveal-to-confirmation window, and nothing in this work makes Bitcoin quantum-safe today.”

- No consensus change, activation claim, or new cryptographic scheme is proposed; no token, trading product, key material, custody, signing, broadcasting, telemetry, or mainnet flows — ever.

- **No fabricated numbers.** Verification-CPU cost is excluded from today's public model precisely because it cannot be measured without a pinned build; it is funded work, not an asserted figure. Every published number names its source or its command.
- "Audit" for external work means a contracted qualified auditor; this lab's own work is testing and measurement, named as such. "Finding" is graded honestly — two real gaps, two open questions, zero consensus breaks.
- Coordinated disclosure: pin and reproduce → minimize → notify maintainers privately → remediation window → publish facts → retest. If BIP 360 is revised, the pinned corpus and cost model retarget the successor revision; if it is abandoned, you stop at the next gate.

8 The applicant, plainly

Ashwin Goyal — B.Tech computer science undergraduate, VIT Bhopal, India; open-source engineer, ~90 public repositories. No Bitcoin Core commits, no published cryptography research, one prior \$200 developer microgrant — stated so you never discount for a claim not made. The structure makes those absences immaterial: you never pay ahead of verified work, and the verified work — a reproducible cost model, an independent differential audit with real findings, a pinned conformance suite — already exists at the link above. Full-time single-owner focus is the delivery model, the public commit history is the accountability model, the gates are the enforcement model.

9 Later phases — earned, not assumed

If this phase delivers, a separately gated proposal (USD 150,000–300,000) would grow the exposure classifier into production custodian migration tooling, add descriptor/PSBT preflight (regtest/signet only), and deepen the cost research into a full post-quantum migration reference architecture with external review. None of that is part of this ask; this phase stands complete on its own — the cost paper, the findings, and the corpus retain their value regardless of what follows.

The model is public, the suite is green, the findings are real, the command is one line. Run it, sample the corpus, forward this to a maintainer — this application welcomes every form of scrutiny you can apply, because scrutiny and measurement are the product it asks you to fund.

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Galaxy Bitcoin Quantum Readiness Initiative · August 2026 · No confidential material — share freely, including with implementation maintainers for diligence. Archived in the repository at [docs/galaxy-proposal.pdf](#).